

Regression Analysis in Machine learning

Regression analysis is a statistical method to model **the relationship between a dependent (target) and independent (predictor) variables** with one or more independent variables. More specifically, Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed. Regression is a **supervised learning technique** which helps in finding the correlation variables and enables us to predict the continuous output variable based on the one or more predictor variables. It is mainly used for **prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables**.

Example: Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

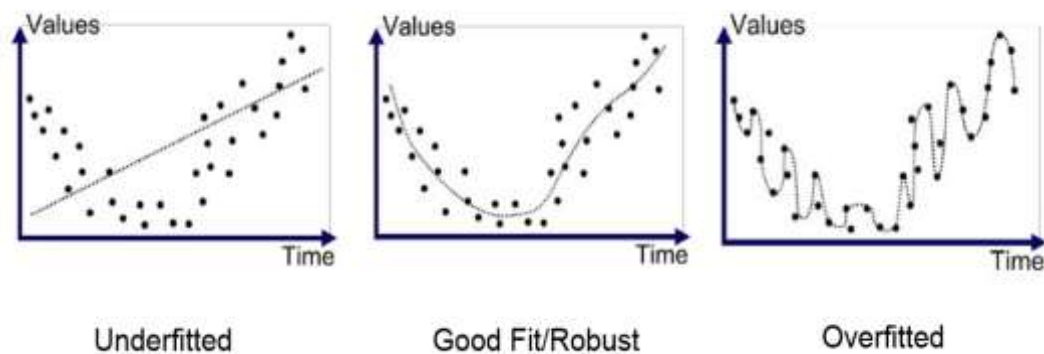
Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??

Now, the company wants to do the advertisement of \$200 in the year 2019 and **wants to know the prediction about the sales for this year**. So to solve such type of prediction problems in machine learning, we need regression analysis.

Terminologies Related to the Regression Analysis:

- **Dependent Variable:** The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.
- **Independent Variable:** The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.
- **Outliers:** Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.
- **Multicollinearity:** If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.
- **Underfitting and Overfitting:** If our algorithm works well with the training dataset but not well with test dataset, then such problem is called Overfitting. And if our

algorithm does not perform well even with training dataset, then such problem is called underfitting.



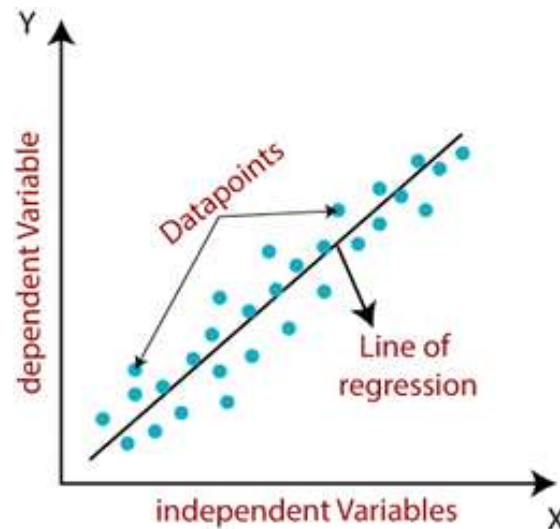
Types of Regression

There are various types of regressions which are used in data science and machine learning. Each type has its own importance on different scenarios, but at the core, all the regression methods analyze the effect of the independent variable on dependent variables. Here we are discussing some important types of regression which are given below:

1. Linear Regression
2. Logistic Regression
3. Polynomial Regression
4. Ridge Regression
5. Lasso Regression

Linear Regression

- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called simple linear regression. And if there is more than one input variable, then such linear regression is called multiple linear regression.
- The linear regression model provides a sloped straight line representing the relationship between the variables. Consider the below image:



Mathematically, we can represent a linear regression as:

$$y = a_0 + a_1x + \varepsilon$$

Here,

Y = Dependent Variable (Target Variable)

X = Independent Variable (predictor Variable)

a_0 = intercept of the line (Gives an additional degree of freedom)

a_1 = Linear regression coefficient (scale factor to each input value).

ε = random error.

The Linear Regression Equation

Linear regression is a way to model the relationship between two variables. You might also recognize the equation as the slope formula. The equation has the form $Y = a + bX$, where Y is the dependent variable (that's the variable that goes on the Y axis), X is the independent variable (i.e. it is plotted on the X axis), b is the slope of the line and a is the y-intercept.

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Example: You are given a dataset containing the Age (in years) and corresponding Glucose Level (in mg/dL) of 6 subjects. The dataset is as follows:

Subject	Age (x)	Glucose Level (y)
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

Using the linear regression model based on the dataset of Age and Glucose Levels, predict the glucose level for a person who is 67 years old.

Solution:

Step 1: Calculate the necessary summations

Subject	Age x	Glucose Level y	xy	x ²	y ²
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

From the above table, $\Sigma x = 247$, $\Sigma y = 486$, $\Sigma xy = 20485$, $\Sigma x^2 = 11409$, $\Sigma y^2 = 40022$. n is the sample size (6, in our case).

Step 2: Use the following equations to find a and b.

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$a = 65.1416$$

$$b = 0.385225$$

Find a:

$$((486 \times 11,409) - ((247 \times 20,485)) / 6 (11,409) - 2472)$$

$$= 484979 / 7445$$

$$= 65.14$$

Find b:

$$(6(20,485) - (247 \times 486)) / (6(11409) - 2472)$$

$$= (122,910 - 120,042) / 68,454 - 2472$$

$$= 2,868 / 7,445$$

$$= 0.385225$$

Step 3: Insert the values into the equation.

$$y = a + bx$$

$$Y = 65.14 + .385225X$$

Step 4: Predict the glucose level for a 67-year-old

To predict the glucose level for a person with age $x = 67$,

$$y = 65.14 + 0.39 \times 67$$

The predicted glucose level for a person who is 67 years old is approximately 90.95 mg/dL.

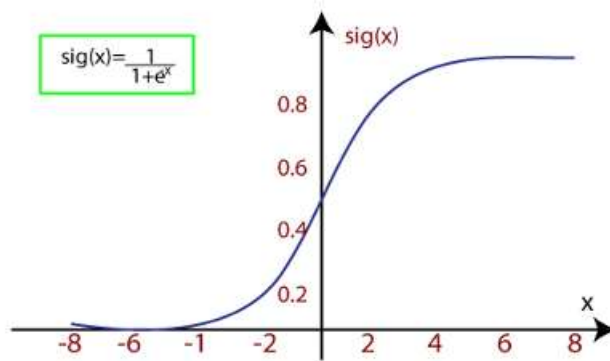
Logistic Regression

- Logistic regression is another supervised learning algorithm which is used to solve the classification problems. In classification problems, we have dependent variables in a binary or discrete format such as 0 or 1.
- Logistic regression algorithm works with the categorical variable such as 0 or 1, Yes or No, True or False, Spam or not spam, etc.
- It is a predictive analysis algorithm which works on the concept of probability.
- Logistic regression is a type of regression, but it is different from the linear regression algorithm in the term how they are used.
- Logistic regression uses sigmoid function or logistic function which is a complex cost function. This sigmoid function is used to model the data in logistic regression. The function can be represented as:

$$f(x) = \frac{1}{1+e^{-x}}$$

- $f(x)$ = Output between the 0 and 1 value.
- x = input to the function
- e = base of natural logarithm.

When we provide the input values (data) to the function, it gives the S-curve as follows:



- It uses the concept of threshold levels, values above the threshold level are rounded up to 1, and values below the threshold level are rounded up to 0.

Example: Given the dataset:

Student	Hours Studied (x)	Passed (y)
1	1	0
2	2	0
3	3	0
4	4	0
5	5	1
6	6	1

Using the logistic regression model derived from the given dataset, predict whether a student will pass or fail based on their study hours, a student who studied 4.5 hours. Where, $\beta_0 = -4.96$ is the intercept, $\beta_1 = 1.08$ is the coefficient for hours studied.

Solution:

Step 1: Logistic Regression Model Equation

We have the logistic regression model equation:

$$P(y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Step 2: We want to predict whether a student will pass if they studied 4.5 hours.

Substitute $x=4.5$, $\beta_0 = -4.96$, $\beta_1 = 1.08$ into the logistic regression equation:

$$P(y = 1) = \frac{1}{1 + e^{-(-4.96 + 1.08 \cdot 4.5)}}$$

Step 3: Calculate the expression inside the exponent

$$-4.96 + 1.08 \cdot 4.5 = -4.96 + 4.86 = -0.10$$

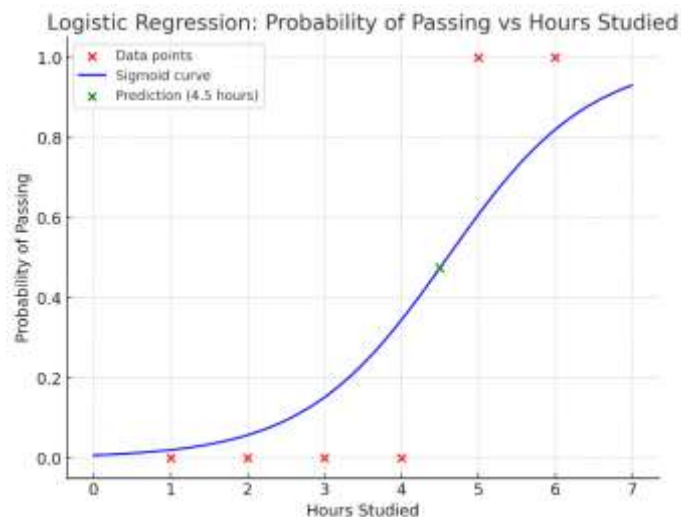
Step 4: Calculate the sigmoid function

$$P(y = 1) = \frac{1}{1 + e^{0.10}}$$

The predicted probability of passing for a student who studied 4.5 hours is approximately 0.475 (or 47.5%).

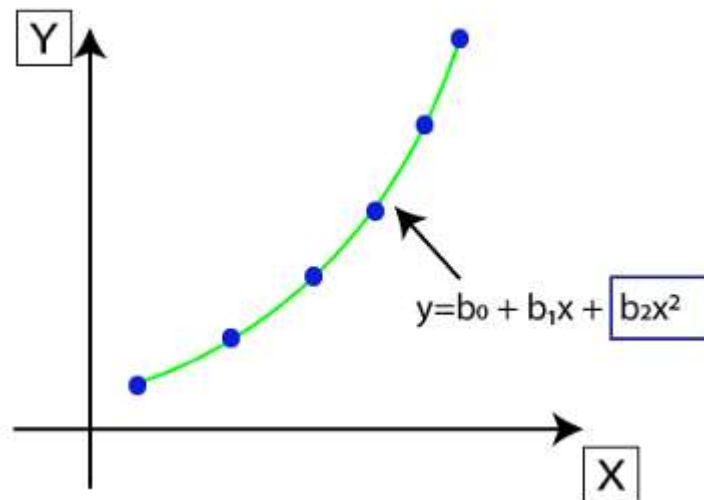
Step 5: Since the probability $P(y=1)$ is less than 0.5, we predict that the student will **fail** (class 0).

Step 6: Plot the point on the graph,



Polynomial Regression

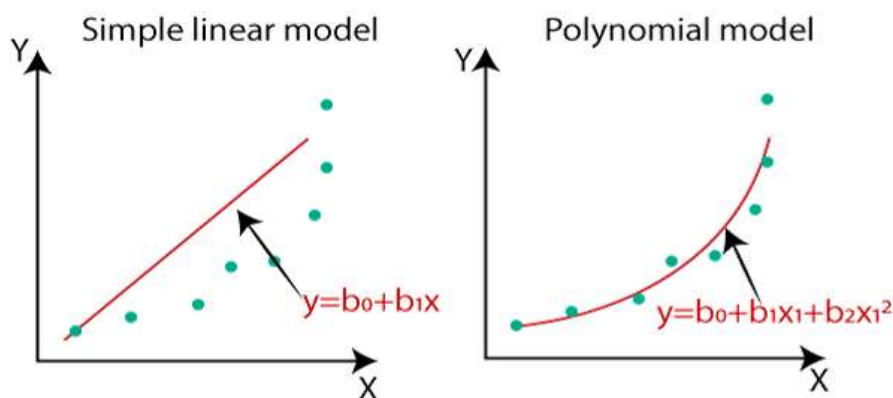
- Polynomial Regression is a type of regression which models the non-linear dataset using a linear model.
- It is similar to multiple linear regression, but it fits a non-linear curve between the value of x and corresponding conditional values of y .
- Suppose there is a dataset which consists of datapoints which are present in a non-linear fashion, so for such case, linear regression will not best fit to those datapoints. To cover such datapoints, we need Polynomial regression.
- In Polynomial regression, the original features are transformed into polynomial features of given degree and then modeled using a linear model.



- The equation for polynomial regression also derived from linear regression equation that means Linear regression equation $Y = b_0 + b_1x$, is transformed into Polynomial regression equation $Y = b_0 + b_1x + b_2x^2 + b_3x^3 + \dots + b_nx^n$.
- Here Y is the predicted/target output, $b_0, b_1, \dots, b_n$ are the regression coefficients. x is our independent/input variable.
- The model is still linear as the coefficients are still linear with quadratic.

Need for Polynomial Regression:

1. If we apply a linear model on a linear dataset, then it provides us a good result as we have seen in Simple Linear Regression, but if we apply the same model without a modification on a non-linear dataset, then it will produce a drastic output. Due to which loss function will increase, the error rate will be high, and accuracy will be decreased.
2. So, for such cases, where data points are arranged in a non-linear fashion, we need the Polynomial Regression model. We can understand it in a better way using the below comparison diagram of the linear dataset and non-linear dataset.



Ridge Regression

- Ridge regression is one of the most robust versions of linear regression in which a small amount of bias is introduced so that we can get better long term predictions.
- The amount of bias added to the model is known as Ridge Regression penalty.
- We can compute this penalty term by multiplying with the lambda to the squared weight of each individual features.
- The equation for ridge regression will be:

$$L(x, y) = \text{Min}(\sum_{i=1}^n (y_i - w_i x_i)^2 + \lambda \sum_{i=1}^n (w_i)^2)$$

- A general linear or polynomial regression will fail if there is high collinearity between the independent variables, so to solve such problems, Ridge regression can be used.
- Ridge regression is a regularization technique, which is used to reduce the complexity of the model. It is also called as L2 regularization.
- It helps to solve the problems if we have more parameters than samples.

Lasso Regression

- Lasso regression is another regularization technique to reduce the complexity of the model.
- It is similar to the Ridge Regression except that penalty term contains only the absolute weights instead of a square of weights.
- Since it takes absolute values, hence, it can shrink the slope to 0, whereas Ridge Regression can only shrink it near to 0.
- It is also called as L1 regularization. The equation for Lasso regression will be:

$$L(x, y) = \text{Min}(\sum_{i=1}^n (y_i - w_i x_i)^2 + \lambda \sum_{i=1}^n |w_i|)$$

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.
In Linear regression, we predict the value of continuous variables.	In logistic Regression, we predict the values of categorical variables.
In linear regression, we find the best fit line, by which we can easily predict the output.	In Logistic Regression, we find the S-curve by which we can classify the samples.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
The output for Linear Regression must be a continuous value, such as price, age, etc.	The output of Logistic Regression must be a Categorical value such as 0 or 1, Yes or No, etc.
In Linear regression, it is required that relationship between dependent variable and independent variable must be linear.	In Logistic regression, it is not required to have the linear relationship between the dependent and independent variable.
In linear regression, there may be collinearity between the independent variables.	In logistic regression, there should not be collinearity between the independent variable.

S.No	Lasso Regression	Ridge Regression
1	Panelizes the sum of absolute value of weights.	penalizes the sum of square weights.
2	It has a sparse solution.	It has a non-sparse solution.
3	It gives multiple solutions.	It has only one solution.
4	Constructed in feature selection.	No feature selection.
5	Robust to outliers.	Not robust to outliers.
6	It generates simple and interpretable models.	It gives more accurate predictions when the output variable is the function of whole input variables.
7	Unable to learn complex data patterns.	Able to learn complex data patterns.
8	Computationally inefficient over non-sparse conditions.	Computationally efficient because of having analytical solutions.